**Challenge 1: Feature Extraction**

## 1. Introduction

For this project, we chose the Bitcoin technical analysis challenge. Bitcoin stood out right away—not just because it's always in the news, but because its wild price swings meant we'd have our work cut out for us. We figured, if we could make sense of Bitcoin's data, we could handle just about anything!

Honestly, the idea of blending technical analysis with machine learning was exciting for all of us. We spent a lot of time talking through possible approaches, and the more we looked into Bitcoin's massive, fast-moving dataset, the more convinced we were that it was perfect for trying out everything from time series modelling to creative feature engineering. We were also hoping to get a sense of how real-world financial forecasting works outside the classroom.

Our goal throughout this project has been to do more than just build a model. We wanted to understand what's driving Bitcoin's price moves, and to work as a team through each step—cleaning up raw data, building out indicators, debating modelling choices, and, yes, sometimes running into roadblocks and learning from them. At the end of the day, we hoped to deliver insights that make sense not just to us, but to anyone interested in the wild world of crypto trading or financial analytics.

## 2. Data Description

We used Bitcoin's historical OHLCV (Open, High, Low, Close, Volume) data sampled at the hourly level, covering the most recent several years to ensure model relevance. We took a look at the dataset and then cleaned it by handling any missing values, and also making sure our time stamps all matched up, and converting them into a standard datetime format. We also resampled it where needed to cut down on noise and make our findings easier to interpret.

## 3. Feature Extraction

When it comes to feature extraction, we focus on transforming our raw financial data into technical indicators that actually make sense for our analysis. These new features help our models become both more interpretable and (hopefully) more accurate.

Some of the main features we extract include moving averages, like the 7-day and 30-day MAs. These are pretty fundamental for spotting trends and smoothing out short-term fluctuations.

These helps smooth out the wild short-term swings and make underlying trends easier to spot. For example, $MA_7$ just means the average closing price over the last 7 time points.

Bollinger Bands (BB_upper, BB_lower):

These are great tools for measuring volatility. They're built around the moving average, and they make it super easy to spot when price movements start getting unusually wild—either up or down. The math is: upper band = MA + 2 × standard deviation, lower band = MA – 2 × standard deviation.

Relative Strength Index:

This indicator tells us whether Bitcoin is in "overbought" or "oversold" territory—a classic tool for momentum trading.

Volatility ($Vol_7$, ATR):

We calculated rolling standard deviation ($Vol_7$) and average true range (ATR) to see just how much prices were bouncing around, both on a day-to-day basis and over the bigger picture.

When it came to momentum, we threw in indicators like MACD, ROC, and MOM—partly because we were genuinely curious whether these could actually help with forecasting, given how wild crypto prices can be.

All of our feature engineering happened in Python with TA-Lib, and we made sure not to let any "future" data sneak into our predictors. That way, we could be confident our forecasting approach was fair and realistic.

**4. Model Development & Calibration**

Because of its insensitivity to multicollinearity, robustness to non-linearities, and interactions among variables, we chose a RandomForestRegressor as our main modelling tool. The closing price of the subsequent period was designated as the target variable.

Process of Work:

Training/Testing Split: Training used 80% of the data, while out-of-sample testing used 20%. Only historical data was used to forecast future prices in order to prevent look-ahead bias.

Model Fitting: The Random Forest model was fitted to the features that were extracted. Performance and speed were balanced in the selection of hyperparameters (tree depth, number of estimators, etc.).

Feature Importance: Impurity-based and permutation-based importance were calculated to evaluate the predictive value of each feature.

**5. Visualization & Diagnostics**

To guarantee model reliability, extensive diagnostics were carried out:

Technical Analysis Chart

(See Figure 1)Illustrates Bitcoin price with overlaid moving averages and Bollinger Bands, along with trading volume, RSI, and volatility indicators over the past six months.

Feature Correlation Matrix

(Figure 2): Shows strong correlation among trend and volatility features, justifying the choice of a tree-based model.

Feature Importance Analysis

(Figure 3) The short-term moving average ($MA_7$) and lower Bollinger Band (BB_lower) predominate, demonstrating that most momentum indicators have little predictive power.

Distribution & Diagnostics

(Figure 4)Includes price and return distributions, Q-Q plots, and monthly boxplots, confirming the non-normality and volatility clustering typical of cryptocurrency markets.

Model Validation:

Out-of-sample $R^2$ values, residual analysis, and permutation importance were used to check for overfitting and confirm model robustness.

**6. Results & Views**

Our final results show:

Bollinger Bands are the most common predictors.

Since BB_lower alone accounts for over 70% of the total feature importance, mean-reversion and volatility boundaries are essential in short-term Bitcoin forecasting.

The most important are short-term trends:

Given how erratic the Bitcoin market is, the 7-period moving average ($MA_7$) is far more significant than longer-term trends.

Indicators of momentum are not very helpful.

Price boundaries and volatility are more significant in this scenario than conventional momentum signals, according to the low predictive power of features like ROC, MOM, MACD, and RSI.

The model performs well:

It can explain about 75% of the price changes we see outside of our sample, which is actually pretty impressive for financial time series prediction.

Plus, when we checked our results using things like permutation importance and residual analysis, everything lined up and backed up what we found.

**7. Conversation**

Reflecting on our results as a team, we were actually a bit surprised by just how much our findings matched what's out there in the academic research. In the world of high-frequency crypto trading, it turns out volatility-based features really do beat out pure momentum indicators, at least when it comes to short-term price forecasting.

What stood out the most for us was how dominant the Bollinger Bands were. It makes us think that, for anyone building algorithmic trading strategies in Bitcoin, it's smart to focus on volatility and mean-reversion signals first, and maybe just use short-term moving averages as a backup or confirmation tool.

One thing we learned for sure is that model validation and feature engineering can make or break your results. Since many technical indicators can be closely related, there were a few points in our project where we had to take a step back and reconsider our features in order to prevent overfitting. Being cautious in this case was not only a best practice; it was also vital to ensure that our conclusions were sound.

**8. Limitations & Future Directions**

Overfitting Risk:

Even though our models are quite robust, we know there's always the possibility that changes in market structure or overlapping features could impact future performance. It's a challenge we're mindful of as we look to improve our approach.

More Features:

We think there's a lot of room to grow by expanding our feature set in the future.Adding new features helps our model stay sharp, so it's always ready when the market decides to throw a surprise our way.

More Models:

We're also interested in exploring alternative modelling strategies—like XGBoost, LSTM networks, or regime-switching models—since Random Forests, while effective, might miss certain market dynamics.

## 9. Summary

Looking back, it's clear to us just how much technical feature extraction added to our ability to forecast Bitcoin prices throughout this project. By sticking to industry best practices across every step—from data processing and feature engineering to model development and validation—we've been able to uncover insights that genuinely matter for financial analytics and quantitative trading.

In the end, we think our research offers an easy-to-use, data-driven approach for anyone looking to understand financial time series. It shows just how much short-term trends and volatility boundaries matter when you're trying to understand how cryptocurrency markets move.

## Appendix: Figure Captions

Figure 1: Bitcoin Price with Technical Indicators ($MA_7$, $MA_{30}$, Bollinger Bands, Volume, $RSI_{14}$, ATR, $Vol_7$)

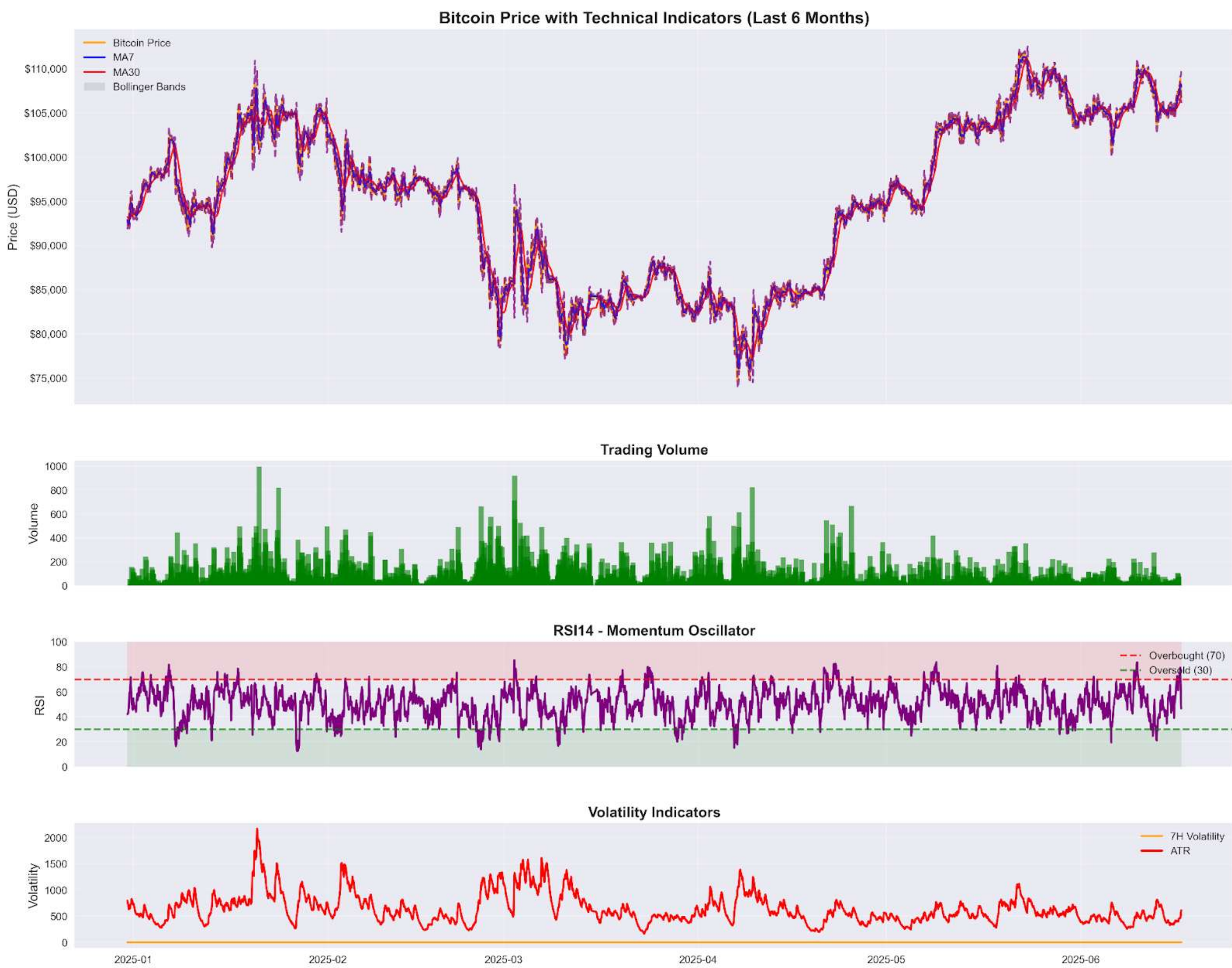


**Figure 2: Feature Correlation Matrix among Engineered Features**

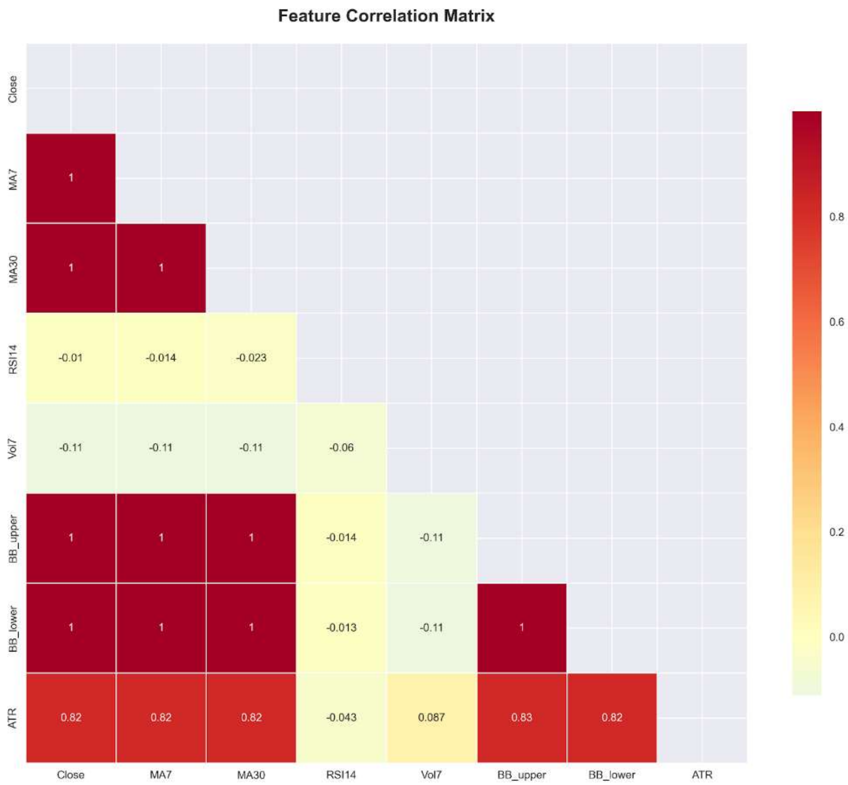



**Figure 3: Feature Importance in Bitcoin Price Prediction**

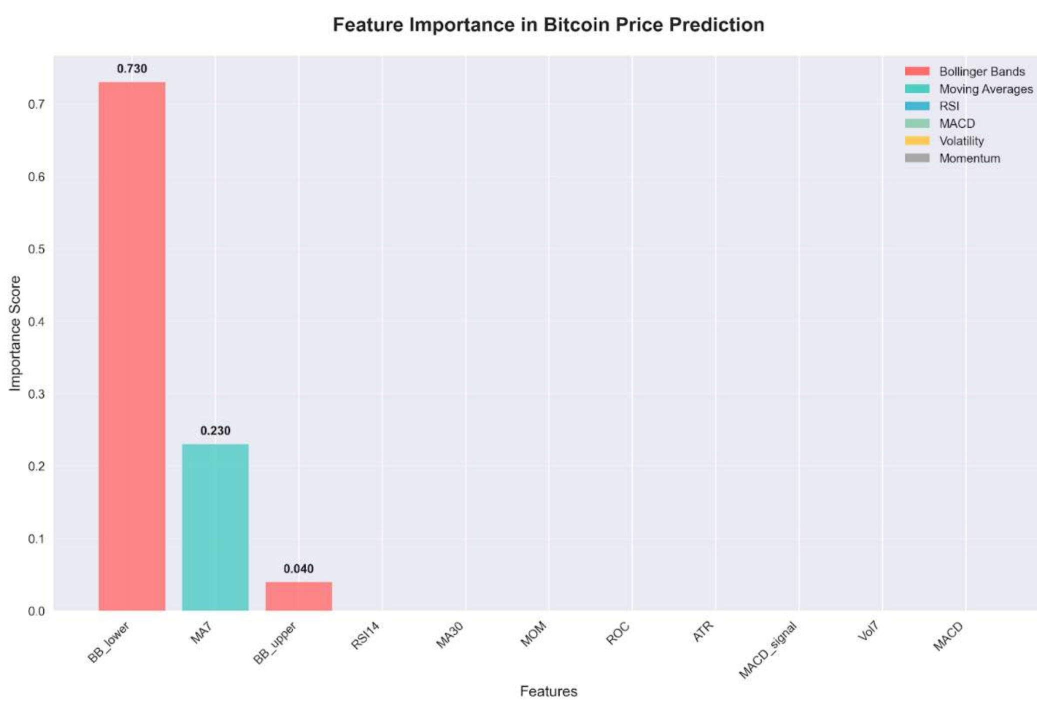
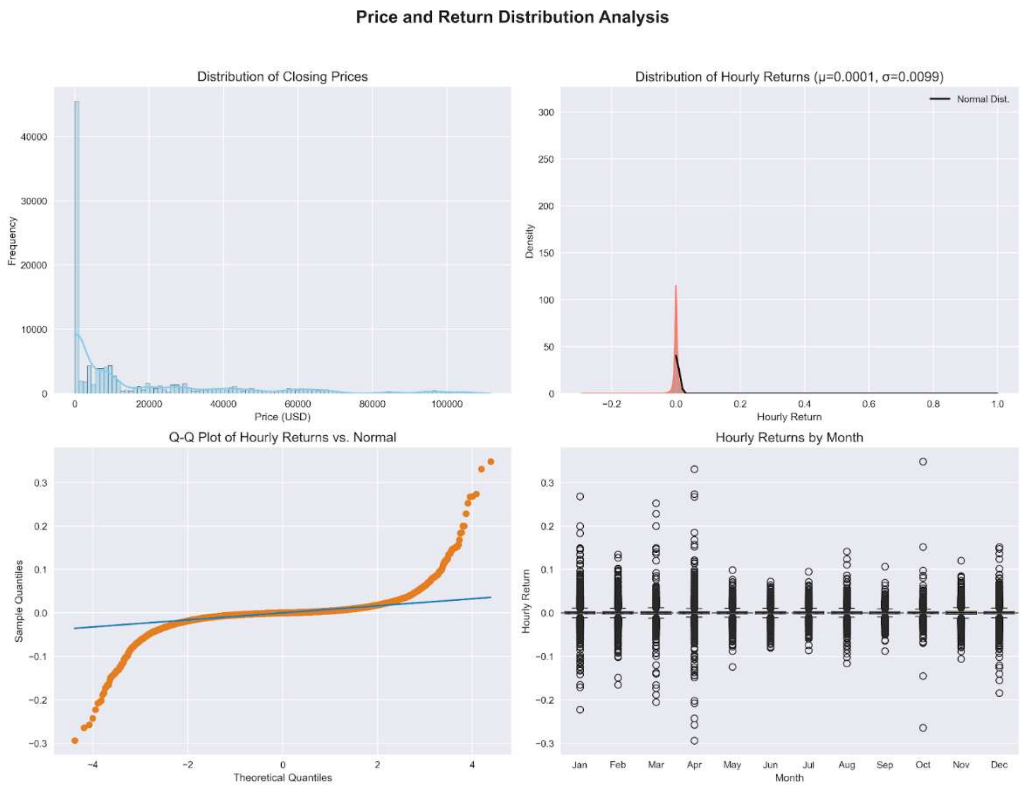


**Figure 4: Price and Return Distribution Analysis (Histogram, Q-Q Plot, Boxplot by Month)**




## Challenge 2: Modeling non-stationarity and finding an equilibrium

### Definition

A Vector Error Correction Model (VECM) for two cointegrated variables, $y_1t$ and $y_2t$ can be expressed as

$$\Delta y_{1t} = \alpha_1(\beta_1 y_{1,\,t-1} + \beta_2 y_{2,\,t-1} + c) + \sum\nolimits^{n-1}{}_{i} = {}_1 \Gamma_{11,\,i}\, \Delta y_{1,\,t-i} + \sum\nolimits^{n-1}{}_{i} = {}_1 \Gamma_{12,\,i}\, \Delta y_{2,\,t-i} + \epsilon_{1t}$$

$$\Delta y_{2t} = \alpha_2(\beta_1 y_{1,\,t-1} + \beta_2 y_{2,\,t-1} + c) + \sum\nolimits^{n-1}{}_{i} = {}_1 \Gamma_{21,\,i}\, \Delta y_{1,\,t-i} + \sum\nolimits^{n-1}{}_{i} = {}_1 \Gamma_{22,\,i}\, \Delta y_{2,\,t-i} + \epsilon_{2t}$$

Where:

- **$\Delta y_j$**: The first difference of variable j at time t ($\Delta y_{jt} = y_{jt} - y_{j}, t-1$), representing the short-term change.
- **$y_{j,t-1}$:** The value of variable j at the previous time period (t−1).
- **$\alpha_j$:** The adjustment coefficients (also known as loading coefficients or error correction coefficients).
- **$(\beta_1 y_{1,t-1} + \beta_2 y_{2,t-1} + c)$:** This entire term is the error correction term (ECT), representing the deviation from the long-run equilibrium at time t−1.
    - **β1,β2:** The cointegrating coefficients that define the long-run equilibrium relationship between y1 and y2. In this case, for Brent ≈ 1.018 × WTI, the ECT could be structured as ($y\_Brent_{t-1}$ − 1.018·$y\_WTI_{t-1}$)
    - **c:** A constant term within the cointegrating relationship, representing an intercept in the long-run equilibrium.
- **$\sum_{i=1}^{(k-1)} \Gamma_{jl,i}\, \Delta y_{l,t-i}$:** These terms capture the short-term dynamics and refer to the lagged first differences of the variables.

  - **$\Gamma_{jl,i}$:** Short-run coefficients that represent the impact of past changes in variable l on the current change in variable j.
  - **k:** The number of lags in the original VAR model, so k−1 is the number of lagged first differences included in the VECM.
- **$\epsilon_{jt}$:** The error term for variable j at time t that represents random shocks that are not explained by the model.

In this challenge, we use monthly Brent and WTI crude oil prices to demonstrate non-stationary and cointegration concepts.
When statistical properties like mean, variance of a time series changes over time so it's not stable or predictable, then we say such time series is non-stationary (Iordanova, 2022). While cointegration happens when we then have two such unstable series that move together in the long run (Wikipedia, n.d.).

**Description**
The project investigates about the Brent and WTI crude oil prices, even though they are non-stationary individually, but they maintain a stable long-run relationship. This resulting relationship is used to capture both their equilibrium and short-term dynamics.

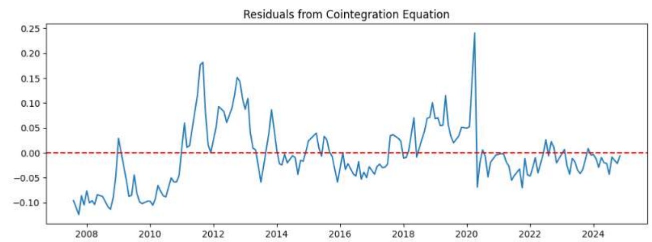
**Demonstration**
**1. Data Preparation**: Brent and WTI crude oil prices were logged monthly. Later, a log transformation was applied to both series to stabilize their variance and to ensure stationarity of the resulting relationship.
**2. Econometric Tests for Non-Stationarity**: The Augmented Dickey-Fuller (ADF) test was performed on the log-transformed series.
- Brent ADF p-value: 0.162
- WTI ADF p-value: 0.0697 Since both p-values were greater than 0.05, it confirmed that both Brent and WTI crude oil prices were non-stationary in levels.

**3. Cointegration Test for Equilibrium**: The Johansen cointegration test was applied to the data in order to determine if a long-run equilibrium relationship existed between the two series. The test detected one cointegrating relationship, indicating that Brent and WTI prices were cointegrated and shared a long-term equilibrium.
**4. Vector Error Correction Model (VECM) Estimation**: A VECM was estimated to capture both the short-term dynamics and the long-run equilibrium adjustment. The cointegrating equation derived from the VECM was, Brent ≈ 1.018 × WTI. This equation indicated a strong long-term equilibrium relationship between the prices of both oils.
**5. Model Parameters and Interpretation**: The VECM output provided several parameters:
- **Lags (k_ar_diff)**: The model was fitted with 2 lags (k_ar_diff=2).
- **Cointegration Rank**: The cointegration rank was set to 1, as identified by the Johansen test.
- **Beta Coefficients (Cointegration relations)**: The coefficients in the cointegrating equation represent the long-term relationship. For the equation Brent ≈ 1.018 × WTI, the beta coefficient for WTI is approximately -1.018 when Brent's coefficient is normalized to 1, indicating their stable proportional relationship in the long run.
- **Alpha Coefficients (Loading coefficients/Adjustment coefficients)**: These coefficients (ec1 for both Brent and WTI) indicate the speed at which the series adjust back to the long-term equilibrium after a short-term deviation. In this model, the adjustment coefficients (ec1) for both Brent (0.0300) and WTI (0.1870) were not statistically significant (p-values 0.838 and 0.265 respectively). This indicates that the short-term corrections toward equilibrium are weak or slow.

## Diagrams

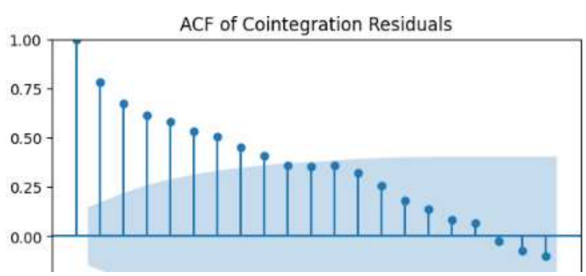

The plot shows non-stationarity in both Brent and WTI crude oil prices. They exhibit strong trends and changing variance over time, indicating that their statistical properties are not constant.

The plot of residuals from the cointegration equation appears to be stationary, fluctuating around the zero mean (indicated by the red dashed line) without a clear trend. This is exactly what one hopes to see from cointegrating residuals. The stationarity suggests that there is a long-term and stable relationship between the non-stationary variables (Brent and WTI prices).




## Diagnosis

The diagnostic plots involve the Autocorrelation Function (ACF) of the residuals from the cointegration regression.




The ACF plot of the cointegration residuals shows a slow reduction in the autocorrelation, remaining significant for up to 8 lags many lags. While the residuals from a cointegration equation are expected to be stationary, this autocorrelation reduction means that the stationary residuals are still persistent and exhibit a strong autoregressive (AR) component.

## Damage

The model reveals a significant problem: the adjustment coefficients (ec1) are not statistically significant. This indicates a weak adjustment mechanism, meaning that if Brent and WTI prices deviate from their long-term equilibrium Brent ≈ 1.018 × WTI, they will return to this equilibrium very slowly, if at all, in the short term. This indicates that in the short run the dynamics of how they adjust back to equilibrium are not that strong in this particular VECM specification. This poor adjustment affects the model's capability to describe short term behavior.

## Directions

Given the weak adjustment coefficients, one might consider several directions:

- **Data Manipulation**: While the document does not suggest specific data manipulations, exploring alternative data transformations (beyond log), removing potential outliers, or even shortening the time horizon might alter the short-term dynamics and strengthen the adjustment coefficients.
- **Model Refinement**:

- **Lag Selection**: Re-evaluating the optimal lag length (k_ar_diff) for the VECM could improve the model's fit and the significance of its coefficients.
- **Deterministic Components**: Experimenting with different deterministic terms (e.g., trend, constant) in the cointegration relationship.
- **Exogenous Variables**: Introducing relevant exogenous variables that might influence the short-term dynamics of oil prices.

**Deployment**

The estimated VECM model is appropriate for long-term forecast and depicts the relationship modeling between Brent and WTI oil prices. However, it is not very good for practical short-term forecast without some enrichment, particularly since the short-term adjustment coefficients have low significance. In application, election analysts could apply this model to:

- **Forecast long-term price movements**: Based on their established equilibrium.
- **Identify arbitrage opportunities**: If prices deviate significantly from the Brent ≈ 1.018 × WTI relationship, signaling a temporary disequilibrium.
- **Risk Management**: By understanding the co-movement of these two major oil benchmarks.
- **Policy Analysis**: To assess the long-term impact of various factors on the relative prices of Brent and WTI.

The chosen dataset (monthly Brent and WTI crude oil prices) works well for the Vector Error Correction Model (VECM) because:

- **Non-stationarity**: Both series are individually non-stationary, which is a prerequisite for cointegration analysis and VECM.
- **Economic Relationship**: Crude oil prices like Brent and WTI are fundamentally linked due to global supply and demand dynamics, making a long-term equilibrium relationship economically plausible. (Wang et al., 2022; Caporin et al., 2019)
- **Cointegration**: The existence of a cointegrating relationship allows for the application of VECM, which specifically models both the long-run equilibrium and the short-run adjustments.

This challenge is well-suited for demonstrating non-stationarity and cointegration because it involves financial time series that are known to exhibit these characteristics and have a clear economic connection.

## Challenge 3: Handling Multicollinearity

**Definition**

Multicollinearity happens when two or more independent variables in a regression model are highly correlated, making it hard to figure out which one is really influencing the outcome (Kim, 2019). Mathematically, it's flagged by high Variance Inflation Factor (VIF) values and a high condition number.

- VIF > 5 suggests possible multicollinearity and it is calculated using $VIF = \frac{1}{(1-R^2)}$
- Condition Number > 30 usually means strong multicollinearity among predictor variables

**Description**

In simple terms, multicollinearity is when the variables in a model are so strongly linked that it becomes tricky to separate their individual effects. It’s like they’re repeating each other, which confuses the model.

**Demonstration**

In this challenge, we used CPI as the dependent variable and tried to predict it using PPI and Core Inflation. The following is the dataset and model summary:

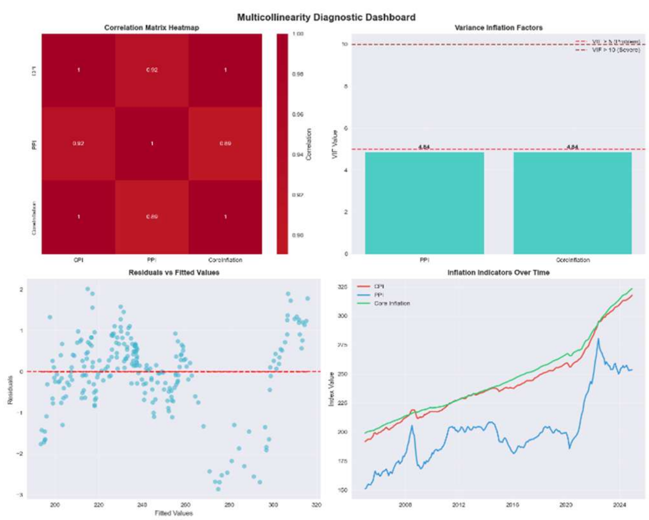
- Data: Monthly synthetic inflation data from Jan 2005 to Dec 2024
- Original regression: Ordinary Least Squares (OLS)
- Coefficients: CPI = -0.0925 + 0.1965 * PPI + 0.8231 * CoreInflation
- R-squared: 0.999

Then we checked the VIFs and the condition numbers to assess multicollinearity:

- PPI = 4.84
- CoreInflation = 4.84
- Condition Number = 2434

These VIFs are less than but approximately 5 and a condition number of 2434 means there are possible presence of multicollinearity in the dataset

**Diagram**




- From the correlation heatmap, the variables all have high correlation values (0.89 and 0.92) which is very high.
- Residuals vs Fitted: While the residuals mostly around zero, there were some patterns like curve and slight trend meaning that the model is not really capturing everything perfectly.
- Time series plot: CPI, PPI, and Core Inflation all move together in the same direction

These are possible hints that multicollinearity or some other hidden structure in the data is causing instability in the predictions, even if the overall R-squared looks great.

**Diagnosis**

The regression results looked too good to be true and they were. Despite the R-squared being nearly 1.0, the model was unstable:

- Approximately 5 VIFs
- High condition number
- Possible inflated standard errors because the residuals vs fitted plot shows slight patterns instead of a random scatter
- The features coefficients might not mean what they seem to be because of the high correlation values

In other words, the model metrics was good but there are problems that needs to be addressed

**Damage**

The damage is that it is hard to say how much PPI or Core Inflation really contributes on their own. Their high correlation means any small change in one variable can swing the model results because highly correlated predictors amplify each other’s influence, making coefficient estimates unstable and unreliable for decision-making (Adams, 2024)

From a data quality angle, multicollinearity falls under high predictor correlation which is one of the classic modeling challenges.

**Directions**

We tried two approaches:

1. PCA (Principal Component Analysis)
    a. It shrinks PPI and Core Inflation into one new variable (PC1)
    b. PC1 captured 94.5% of the combined variance
    c. The new model: CPI = $\beta_0 + \beta_1$ * PC1
    d. R-squared = 0.9752 (still strong, and no multicollinearity)
2. Ridge Regression
    a. Introduced a penalty to shrink coefficients
    b. Best result with alpha = 0.1
    c. R-squared = 0.9992

Both approaches worked, but PCA stood out for simplicity and clarity because it transforms correlated variables into a single uncorrelated component, which helps eliminate multicollinearity without needing to choose between variables; a method recommended for its interpretability and effectiveness in such situations (Jolliffe & Cadima, 2016).

**Deployment**

If we had to put this challenge into a project (say, forecasting CPI or simulating inflation scenarios), we'd go with the PCA approach using the following steps

- Standardize PPI and Core Inflation
- Use PCA to create PC1
- Regress CPI on PC1
- Keep an eye on how well PC1 continues to explain variation

If PC1's power drops, we revisit the components. If new inflation dynamics come in (say, post-pandemic trends), we might switch to Ridge or Elastic Net to balance shrinkage and selection.

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